

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech. Branch : Electronics and Computer Engineering Semester : III

Subject Code & Name: BTESC304 & Computer Architecture & Operating Systems

Max Marks: 60M

Date: 15/03/2023

Duration: 03:00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12M
A) Describe the structural overview of computer with detail block diagram?	BL2 & CO1	6
B) Differentiate between RISC and CISC architecture? Which one is more powerful and why?	BL2 & CO1	6
C) Explain in detail different types of addressing modes with example of each?	BL2 & CO1	6
Q.2 Solve Any Two of the following.		12M
A) Differentiate between RAM and ROM with respect to devices characteristics?	BL3& CO2	6
B) Explain in detail block diagram of memory management unit (MMU) and explain different functions performed by it?	BL2& CO2	6
C) Write a short note on i) Logical and Physical address mapping in memory ii) Page replacement algorithms	BL3& CO2	6
Q. 3 Solve Any Two of the following.		12M
A) What is the difference between a hardwired implementation and a micro programmed implementation of a control unit design?	BL2& CO3	6
B) With a neat sketch explain the block diagram of Direct memory access (DMA) controller?	BL2& CO3	6
C) Draw and explain basic concept of pipelining? How we can improve the performance of system by using pipelining?	BL1& CO3	6
Q.4 Solve Any Two of the following.		12M
A) Draw and explain in detail inter process communication? State the role of dispatcher and scheduler in inter process communication?	BL1& CO4	6
B) Write a short note on i) System Services ii) Concept of virtual Machines iii) Functions of operating systems	BL2& CO4	6
C) State the scheduling criteria and explain any one scheduling algorithm in details with diagram?	BL3& CO4	6
Q. 5 Solve Any Two of the following.		12M
A) Explain the concept of semaphore? State its types?	BL2& CO5	6
B) What is deadlock? Explain deadlock prevention and deadlock avoidance concept?	BL3& CO5	6

C) Write a short note on i) Deadlock detection ii) Recovery from deadlock

BL1& CO5

6

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